ℓ
ℓ
??
$e^{i\ell\phi}$
$\ell_{,0}$
$\ell\times$
2π
$\phi\in$
$[0,2\pi)$
$5.1\times$
10^4
??
$\ell=$
1
98.66%
$|Overlap|^2$
60.92
$\ell=$
0
53.88
$\ell=$
2
97.96%
$|Overlap|^2$
7.99
normalized shape agreement
$|Overlap|^2$
$\ell,0$
$r_{peak}=$
$w_0\sqrt{\ell/2}$
$4.5\,\mu m$
ℓ
$|Overlap|^2$
$|Overlap|^2$
$\ell\equiv$
$4,5$
82%
??
(i) Azimuthal phase gradient resolution limit:
ℓ
$4.5\,\mu m$
$50\,grid/\mu m$
(ii) Insufficient design domain size relative to mode scale:
$\ell\equiv$
$4,5$
ℓ
??
1%
ℓ
?
?
?

		TiO_2	TiO_2	$\sim\ell\overset{\sim}{\leq}100$		
This work	**Inverse design (topology optimization)**	**GaP all-dielectric**	**82**	**$\sim$ 98 ($\ell\leq3$)**	**Active (WS$_2$**	**Fixe**
					valley-polarized exciton)	

??
(i) Simultaneous high purity and ac-